

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I MOHAMED ASLAM H (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MOHAMED ASLAM H (192572147)

15
—
20

- 1) A smart home automation system is being designed to manage lighting, temperature and security based on user behaviour and environmental conditions. *

Intelligent Agent

It is a system that perceives its environment using sensors and performs actions using actuators to achieve specific goals.

4) Simple reflex Agent

- * Acts only on the current percept
- * Uses predefined condition-action rules.
- * Does not store past information

2) Model-Based Reflex Agent

- * Maintains an internal model of the environment
- * Remembers previous states
- * Makes better decisions than a simple reflex agent

3) Goal Based Agent

- * Makes decisions to achieve predefined goals.
- * Evaluates different possible actions before acting

4

Utility - Based Agent

* selects the action with the highest utility (best overall benefit)

* considers multiple factors simultaneously

Comparison

Agent Type	Memory	Goal	Learning
Simple Reflex	No	No	No
Model Based	yes	No	No
Goal Based	yes	yes	Limited
Utility Based	yes	yes	Can be extended

Best Approach

A hybrid model approach combining Model-Based and Utility Based Agents is the most effective

Reason:-

* Model Based Agent: Remembers user behaviour and environmental conditions

* Utility Based Agent: selects the best actions considering Comfort, Energy efficiency and security.

* The System can also incorporate learning to adapt to user preferences over time

Conclusion: A hybrid Model-Based + Utility Based agent provides the best performance by ensuring user comfort, reducing energy consumption.

2) 1. Uniform Cost Search

19/05/2024

Uniform Cost Search expands the nodes with the lowest cumulative path cost first.

Working:

1. Start from the initial position.
2. Store nodes in a priority queue.
3. Expand the node with the smallest cost.
4. Continue until the end is found.

Advantages:

- * Finds the least cost path.
- * Complete if path costs are positive.
- * Optimal solution.

Disadvantage

- * High memory usage.
- * Slow for very large maps.

Complexity

Time: Depends on branching factors.

Space: Exponential.

2. Iterative Deepening Search (IDS)

IDS combines the advantage of Depth First Search and Breadth First Search (BFS).

Working

1. Search with depth limit 0.
2. Increase depth limit to 1, then 2 and so on.
3. Repeat until the goal is found.

Advantages

- * Very low memory requirement
- * Complete

Disadvantages

- * Repeats node expansions
- * Not suitable for varying path cost

Complexity

* Time: $O(b^d)$

* Space: $O(b^d)$

where:

b = branching factor

d = depth of the solution.

Best Combination and Justification

Justification :

Goal Based agent focuses on reaching the goal
It guarantees the lowest cost path.

Conclusion:

A Goal Based Agent using DFS provides the most reliable and optimal navigation for an unknown maze.